Spectral Optimization Across Vision Architecture Families: A Controlled Empirical Study

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Abstract

SGD achieves 69.50% on ResNet-50 but collapses to 11.51% on ConvNeXt-Tiny under identical training conditions: a 58-point gap caused not by hyperparameter failure but by a fundamental mismatch between optimizer and architecture. We hypothesize that optimizers that explicitly exploit weight-matrix spectral structure should be robust to such architectural variation. Testing this through a controlled study of three spectral optimizers (Muon, SOAP, Shampoo) across three architectures and three datasets, we find: (1) Muon outperforms AdamW by $+5.9$ to $+24.3$ pp across all nine configurations; on CIFAR-100, where all three spectral methods were compared, each outperforms AdamW by $+9$–$+16$ pp with intra-family differences of 1.4–4.1 pp, small relative to the AdamW gap. The narrow intra-family spread establishes spectral optimization as a family-level advantage; (2) the spectral advantage is architecture-dependent, largest on layer-normalized ConvNeXt-T ($+15.0$–$+24.3$ pp) and smallest on batch-normalized ResNet-50 ($+5.9$–$+11.5$ pp). Layer-wise spectral analysis reveals a striking sign reversal in the correlation between spectral divergence and optimizer benefit (ρ=$+0.34$ on CNNs, ρ=-0.66 on transformers). This sign reversal exposes fundamentally different optimization mechanisms; (3) spectral methods reach near-ceiling accuracy within 20 epochs, surpassing 50-epoch AdamW, particularly in short-schedule training regimes. The gap narrows substantially on CNNs with longer training. These findings reframe optimizer selection from hyperparameter tuning to a structural design decision coupled to architectural inductive biases.

1 Introduction

A single architectural change (replacing batch normalization with layer normalization) causes SGD to collapse from 69.50% to 11.51% on the same task (CIFAR-100, identical training protocol). This is not a hyperparameter failure: a five-point learning rate sweep confirms the degradation across the entire viable range (Appendix E). The same shift barely affects spectral optimizers, which achieve 74%+ on both architectures. This stark contrast, where an architectural design choice silently determines whether an optimizer can function, exposes a coupling that current optimizer selection practice does not account for, and motivates a fundamental question: *what properties of an optimization algorithm determine whether it transfers across architectural design choices?*

We hypothesize that the answer lies in how optimizers interact with weight-matrix spectral structure. Modern architectures differ in normalization (batch vs. layer), convolution topology (standard vs. depthwise separable), and attention mechanisms, all of which reshape the spectral properties of weight gradients. *Spectral optimizers*, methods that explicitly exploit this structure through matrix decomposition or Newton–Schulz orthogonalization Bernstein and Newhouse [2024], should therefore be more robust to architectural variation than methods relying on diagonal curvature approximations (AdamW) or raw gradient direction (SGD). This hypothesis is grounded in the theoretical insight of Bernstein & Newhouse Bernstein and Newhouse [2024], who show that Adam, Shampoo, and Muon can each be understood as steepest descent under different matrix norms; our study provides the first layer-level empirical validation of this norm-based optimizer unification across architectures.

We test this hypothesis through a controlled study of three spectral optimizers, namely Muon Jordan et al. [2024][1] (Newton–Schulz orthogonalization), SOAP Vyas et al. [2024] (Kronecker-factored eigenbasis), and Shampoo Gupta et al. [2018] (full-matrix preconditioning), against AdamW Loshchilov and Hutter [2019] and SGD across three architectures (ResNet-50 He et al. [2016], DeiT-Small Touvron et al. [2021], ConvNeXt-Tiny Liu et al. [2022]), three benchmarks (CIFAR-100 Krizhevsky and Hinton [2009], Tiny-ImageNet Le and Yang [2015], ImageNet-100 Deng et al. [2009]), and two training horizons (20 and 50 epochs), with independently tuned learning rates per optimizer–architecture pair (Section 4.2). Our findings:

1. **Family-level spectral advantage.** Muon outperforms AdamW by $+5.9$ to $+24.3$ pp across all nine architecture–dataset configurations; on CIFAR-100, where all three spectral methods were evaluated, each outperforms AdamW by $+9$–$+16$ pp with intra-family differences of 1.4–4.1 pp. The benefit is algorithmic-family-level, not single-method-specific (Section 5.1).

2. **Architecture-dependent magnitude with spectral explanation.** The advantage scales with architectural spectral complexity: largest on layer-normalized ConvNeXt-T ($+15.0$–$+24.3$ pp), smallest on batch-normalized ResNet-50 ($+5.9$–$+11.5$ pp). Layer-wise analysis reveals that per-layer spectral divergence correlates with per-layer optimizer benefit, with opposite sign on CNNs ($\rho{=}{+}0.34$) vs. transformers ($\rho{=}{-}0.66$); these opposite correlations expose distinct optimization mechanisms (Sections 5.2, 5.5).

3. **Convergence efficiency and architecture robustness.** Muon reaches near-ceiling accuracy within 20 epochs, surpassing 50-epoch AdamW with approximately 2.0–$2.4\times$ wall-clock savings after accounting for 5–22% per-iteration overhead (Appendix Table 14), while SGD fails catastrophically on layer-normalized architectures. Spectral methods maintain effectiveness across both normalization paradigms (Sections 5.3, 5.4).

2 Related Work

Optimizer landscape in vision. SGD dominated training throughout the convolutional era He et al. [2016], but Vision Transformers Dosovitskiy et al. [2021], Touvron et al. [2021], Liu et al. [2021] and efficient CNNs Liu et al. [2022], Tan and Le [2019], Howard et al. [2017] shifted practice to AdamW Loshchilov and Hutter [2019] (the decoupled weight decay extension of Adam Kingma and Ba [2015]). Li et al. Li et al. [2024] formalize this entanglement as *backbone–optimizer coupling bias* (BOCB). They benchmark roughly 20 optimizers across diverse backbone families and reveal that optimizer rankings shift substantially with architecture choice. Our work complements BOCB in three ways: (a) we introduce spectral methods (Muon, SOAP, Shampoo) as new baselines that postdate or were absent from the BOCB study; (b) we provide layer-level analysis via a weight-swap protocol and spectral response ratio that expose *where* in the network coupling arises, a granularity BOCB does not address; and (c) we trade breadth for depth, running controlled 3-seed comparisons with statistical significance tests on a focused set of configurations. Prior benchmarks Wilson et al. [2017], Choi et al. [2020], Schmidt et al. [2021] demonstrate that no single optimizer dominates universally, but none systematically control for architecture type across CNN and transformer backbones under matched conditions.

Architecture–optimization interaction. CNNs and ViTs present fundamentally different optimization landscapes: ViTs converge to sharper minima Keskar et al. [2017] with Hessian eigenvalues nearly an order of magnitude larger than those of ResNets Chen et al. [2022], multi-head

[1] At the time of writing, the Muon reference is a blog post and has not been peer-reviewed.

self-attention reshapes the loss surface toward flatter regions Park and Kim [2022], and batch normalization Ioffe and Szegedy [2015] smooths the landscape in ways that layer normalization does not Santurkar et al. [2018], Ba et al. [2016]. Other normalization variants such as Group Normalization Wu and He [2018] and Weight Standardization Qiao et al. [2019] offer alternative smoothing properties but remain less prevalent in modern architectures. SGD performs well with batch normalization but struggles on layer-normalized architectures Liu et al. [2022], a failure mode that adaptive methods overcome.

Spectral and preconditioned methods. Beyond element-wise adaptive methods originating with Adagrad Duchi et al. [2011], K-FAC Martens and Grosse [2015], Shampoo Gupta et al. [2018], Anil et al. [2021], Shi et al. [2023], and SOAP Vyas et al. [2024] exploit gradient matrix structure for preconditioning. Muon Jordan et al. [2024] takes a different approach, directly orthogonalizing the gradient via Newton–Schulz iterations to perform steepest descent under the spectral norm (details in Appendix F). Bernstein & Newhouse Bernstein and Newhouse [2024] prove that Adam, Shampoo, and Muon can each be understood as steepest descent under different matrix norms, providing a theoretical foundation for treating these methods as a *spectral optimizer family*. Our work provides the first layer-level empirical validation of this norm-based unification across architectures. Pethick et al. Pethick et al. [2025] formalize this further via norm-constrained linear minimization oracles, showing that Muon is a special case of their Scion framework. GaLore Zhao et al. [2024] takes a complementary approach, projecting gradients into low-rank subspaces for memory efficiency rather than preconditioning.

Other optimizer directions. Several concurrent lines of work push beyond Adam from complementary angles. Lion Chen et al. [2024] discovers an update rule via symbolic search that uses only the sign of the momentum, achieving competitive performance with lower memory. Sophia Liu et al. [2024] estimates a diagonal Hessian cheaply to scale updates, sharing the second-order motivation of Shampoo but targeting language model pre-training. Adafactor Shazeer and Stern [2018] factorizes the second-moment matrix with a Kronecker decomposition to reduce memory, a design choice echoed in SOAP's later factorization strategy. LARS You et al. [2017] and LAMB You et al. [2020] enable large-batch training via layer-wise adaptive rates, while RAdam Liu et al. [2020] stabilizes Adam's early-training variance without manual warmup. Prodigy Mishchenko and Defazio [2024] eliminates the learning rate hyperparameter entirely through online distance estimation. These methods represent important and active research directions; our experiments focus specifically on spectral and Newton-type preconditioners to enable controlled layer-wise analysis, leaving broader optimizer coverage to future work.

3 Method

3.1 Spectral Optimization via Orthogonalized Updates

Muon Jordan et al. [2024] optimizes 2D weight matrices by orthogonalizing the gradient before applying the update, effectively performing a Newton-like step in the spectral domain. Given a gradient matrix $G \in \mathbb{R}^{m\times n}$ for a weight matrix W, the goal is to compute an approximation to $(GG^\top)^{-1/2}G$, which projects the gradient onto the Stiefel manifold of orthonormal row matrices. Rather than forming this matrix inverse explicitly, Muon uses Newton–Schulz iteration to approximate it efficiently. Starting from $X_0 = G/\|G\|_F$, each iteration applies a quintic polynomial:

$$X_{k+1} = a\,X_k + b\,X_k X_k^\top X_k + c\,X_k(X_k^\top X_k)^2, \tag{1}$$

with fixed coefficients $a = 3.4445$, $b = -4.7750$, $c = 2.0315$ chosen to maximize the convergence basin. Five iterations suffice for convergence in practice (see Appendix F for background on the classical form). The orthogonalized gradient is then used as the update direction, combined with Nesterov momentum ($\beta = 0.95$).

This procedure captures inter-parameter curvature structure that first-order methods miss. SGD updates along the raw gradient direction; AdamW applies element-wise scaling via running second-moment estimates, which amounts to a diagonal approximation to the curvature. Muon's orthogonalization operates on the full gradient matrix, providing a non-diagonal second-order approximation related to K-FAC Martens and Grosse [2015] but without explicitly maintaining Kronecker-factored curvature estimates. As described in Section 4, Muon is applied exclusively to 2D weight matrices (convolution kernels reshaped to 2D and linear layers), while remaining parameters are handled by a separate AdamW instance (Appendix Table 4 details the parameter routing).

3.2 Layer-wise Spectral Analysis Framework

Global accuracy differences between optimizers do not reveal *where* in the network one optimizer outperforms another. We introduce two complementary analysis tools that localize optimizer effects to individual layers.

Weight Swap Protocol. Given a Muon-trained model with weights $\theta^{\text{Muon}} = \{W_l^M\}$ and an AdamW-trained model with weights $\theta^{\text{AdamW}} = \{W_l^A\}$, we construct a hybrid model $\theta^{\text{Muon}\setminus l \leftarrow \text{AdamW}_l}$ by replacing the weights of a single layer l in the Muon model with the corresponding AdamW weights, keeping all other layers at their Muon-trained values. We define the per-layer Muon benefit:

$$\Delta a_l = \text{acc}(\theta^{\text{Muon}}) - \text{acc}(\theta^{\text{Muon}\setminus l \leftarrow \text{AdamW}_l}), \tag{2}$$

where $\text{acc}(\cdot)$ denotes validation accuracy. $\Delta a_l > 0$ indicates that Muon's training at layer l outperforms AdamW; $\Delta a_l < 0$ indicates the reverse. The beneficial layer ratio—the fraction of layers where $\Delta a_l > 0$—aggregates this across all eligible layers.

Spectral Correlation Analysis. To connect the weight swap results with spectral structure, we define what we term the *spectral response ratio* of a weight matrix:

$$R_l = \frac{\sigma_1(W_l)}{\text{sr}(W_l)}, \quad \text{where } \text{sr}(W) = \frac{\|W\|_F^2}{\sigma_1(W)^2} \text{ is the stable rank,} \tag{3}$$

and $\sigma_1(W)$ is the largest singular value. The spectral response ratio captures how the dominant singular value relates to the effective dimensionality of the weight matrix. We choose this ratio over alternatives (effective rank, condition number) because it is bounded, well-defined for rank-deficient matrices, and, crucially, permits direct cross-architecture comparison between convolutional and attention layers without architecture-specific normalization. For convolutional layers, we reshape the 4D kernel tensor to a 2D matrix before computing R_l. We compute the per-layer spectral divergence $\Delta R_l = R_l^{\text{Muon}} - R_l^{\text{AdamW}}$ and evaluate the Spearman rank correlation between ΔR_l and Δa_l (Eq. 2); results are reported in Section 5.5.

3.3 Evaluation Metrics and Statistical Protocol

We report best validation accuracy (early-stopping checkpoint) throughout, in percentage points (pp), to decouple optimizer comparison from convergence speed. Key experiments use 3–4 seeds (from $\{42, 0, 1, 2\}$; Appendix Table 10), reporting mean $\pm$ std (ddof $= 1$). Independent 50-epoch runs with cosine annealing ($T_{\max} = 50$) and the same per-pair learning rates test gap persistence without schedule-length confounding.

4 Experimental Setup

4.1 Datasets and Architectures

We evaluate on three benchmarks: CIFAR-100 Krizhevsky and Hinton [2009] (60K images, 100 classes, 32×32 resized to 224×224), Tiny-ImageNet Le and Yang [2015] (100K images, 200 classes, 64×64), and ImageNet-100 Deng et al. [2009] (a 100-class subset at native 224×224). Training augmentation uses `Resize(256)`, `RandomCrop(224)`, and random horizontal flip.

We select three architectures with comparable parameter counts (22–29M) spanning the major design paradigms: ResNet-50 He et al. [2016] (batch normalization, ∼25.6M), DeiT-S Touvron et al. [2021] (layer normalization, multi-head attention, ∼22M), and ConvNeXt-T Liu et al. [2022] (layer normalization, depthwise separable convolutions, ∼28.6M). All models are trained from scratch with He initialization He et al. [2015] using `timm` Wightman [2019] without pretrained weights.

4.2 Optimization Protocol

We compare five optimizers: Muon Jordan et al. [2024] (spectral-norm orthogonalization via Newton–Schulz iteration, Nesterov β=0.95, applied to 2D weight matrices with a companion AdamW for remaining parameters), SOAP Vyas et al. [2024] (Kronecker-factored eigenbasis preconditioning), Shampoo Gupta et al. [2018] (full-matrix preconditioning), AdamW Loshchilov and Hutter [2019] (β_1=0.9, β_2=0.999, wd 0.05), and SGD Sutskever et al. [2013] (Nesterov momentum 0.9, wd 10^{-4}). All configurations use batch size 128, cosine annealing, and 20 epochs for the main comparison with 50-epoch extensions for convergence analysis.

To ensure fair comparison, each optimizer–architecture–dataset combination receives an independent 5-epoch learning rate sweep over 3–7 log-spaced rates Schmidt et al. [2021]. Of 27 optimizer–architecture–dataset configurations, 18 confirmed a clear peak (accuracy declining at adjacent rates); 7 remained monotonic within the swept range (e.g., Muon on ConvNeXt-T at all three datasets, Shampoo on ResNet-50); and 2 used literature defaults without a formal sweep (Muon on ResNet-50 CIFAR-100, lr=0.02; SGD on ResNet-50, lr=0.1). For monotonic cases, we selected the highest-accuracy learning rate and validated the choice via 20-epoch saturation experiments; further increases either diverged or yielded no meaningful accuracy gain (Appendix Tables 6–8). This conservative protocol means our reported gaps may *underestimate* the true spectral advantage for these configurations, since the optimal learning rate may lie beyond the swept range. Selected learning rates and complete optimizer configurations are reported in Appendix A.

4.3 Evaluation and Reproducibility

We report best validation accuracy across all epochs (i.e., the early-stopping checkpoint) and quantify optimizer gaps in percentage points (pp). Key experiments are replicated with 3 seeds, reporting mean ± std. Independent 50-epoch runs verify whether gaps persist at longer horizons. All results are obtained on a single NVIDIA RTX 5090 GPU with PyTorch and `timm`; code and training logs will be released upon publication.

5 Results

5.1 Main Results: 20-Epoch Cross-Architecture Comparison

If spectral optimizers succeed by exploiting weight-matrix structure, their advantage should be universal across architectures but variable in magnitude, depending on how much exploitable structure each architecture presents. Table 1 confirms this prediction: Muon outperforms AdamW across all nine architecture–dataset pairs, with gaps ranging from +5.9 pp (ResNet-50/Tiny-ImageNet) to +24.3 pp (ConvNeXt-T/ImageNet-100), as visualized in Figure 1. Multi-seed variance is low (standard deviations 0.10–2.17 pp; Appendix Table 10), confirming that observed gaps substantially exceed run-to-run noise.

Spectral Family Comparison on CIFAR-100. A critical question for interpretation is whether the advantage is specific to Muon's Newton–Schulz mechanism or shared across spectral methods. We evaluate SOAP Vyas et al. [2024] and Shampoo Gupta et al. [2018] on CIFAR-100 under the same protocol (Table 1; full comparison in Appendix Table 15). SOAP matches Muon within noise (≤0.4 pp) on ResNet-50 and ConvNeXt-T, and significantly outperforms it on DeiT-S (65.77 ±

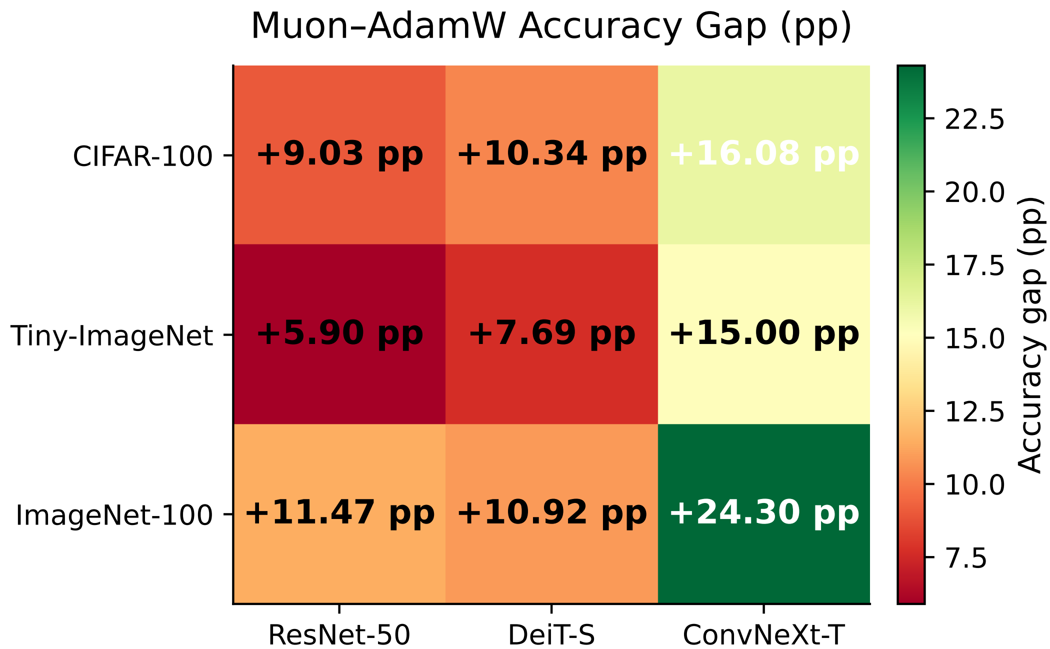

Figure 1: Spectral–AdamW accuracy gap (pp) at 20 epochs. ConvNeXt-T consistently exhibits the largest gap ($+15.0$–$+24.3$ pp); all gaps are positive across the full 3×3 matrix.

0.39% vs. $62.67 \pm 0.41\%$, $+3.10$ pp; Welch's $t \approx 10.2$, $p < 0.001$, significant after Bonferroni correction for 6 intra-family comparisons); on ResNet-50, the difference is not significant ($+0.40$ pp; uncorrected $p{\approx}0.07$). The decisive pattern: all spectral optimizers outperform AdamW by $+9$–$+16$ pp on CIFAR-100, a gap that dwarfs the intra-family differences. This establishes the benefit as a *family-level* property of exploiting weight-matrix spectral structure, not an artifact of any single algorithm's design choices.

Shampoo: Architecture-Dependent Position and LR Sensitivity. After extended learning rate sweeps (Appendix A), Shampoo outperforms AdamW by $+10$–$+12$ pp on all three architectures. A refined sweep for ResNet-50 (from lr$=10^{-2}$ to $5{\times}10^{-2}$) raised Shampoo from 73.84% to $78.25{\pm}0.18\%$, making it the best optimizer on this architecture ($+0.97$ pp over SOAP, $+1.37$ pp over Muon). This $+4.4$ pp gain from a single hyperparameter underscores the acute learning rate sensitivity of full-matrix preconditioning methods. The intra-family ranking is architecture-dependent: Shampoo leads on ResNet-50, SOAP on DeiT-S (65.77% vs. 63.53%, $+2.24$ pp), and Muon on ConvNeXt-T (74.62% vs. $70.57{\pm}0.29\%$, $+4.05$ pp). On DeiT-S, Shampoo outperforms Muon ($+0.86$ pp; Welch's $t{\approx}4.18$, uncorrected $p{\approx}0.023$, adjusted $p{\approx}0.14$ after Bonferroni correction for 6 comparisons; $n_{\text{Shampoo}}{=}3$, $n_{\text{Muon}}{=}4$). This pattern (no single spectral method dominates across architectures) suggests that different preconditioning strategies interact differently with parameter structure, consistent with the sign reversal in spectral correlations (Section 5.5).

5.2 Architecture Dependence of the Spectral–AdamW Gap

If the spectral advantage arises from exploiting weight-matrix structure, its magnitude should vary with how much exploitable structure different architectures present. Figure 1 confirms a consistent ordering: ConvNeXt-T exhibits the largest gap at every dataset ($+16.08$, $+15.00$, $+24.30$ pp), 1.5–$2.5\times$ larger than the other architectures, while ResNet-50 shows the smallest. This ordering is interpretable: ConvNeXt-T's depthwise separable convolutions with layer normalization produce weight matrices with rich spectral structure that diagonal methods cannot exploit, whereas ResNet-

Table 1: Best validation accuracy (%) at 20 epochs. Multi-seed results ($\pm$ std) where available; ConvNeXt-T SGD is single-seed (seed = 42). Bold = best per column. Gap = best spectral − AdamW in pp. SOAP and Shampoo evaluated on CIFAR-100 only.

Dataset	Optimizer	ResNet-50	DeiT-S	ConvNeXt-T
CIFAR-100	Shampoo	**78.25**±0.18	63.53±0.05	70.57±0.29
	SOAP	77.28±0.20	**65.77**±0.39	74.40±0.43
	Muon	76.88±0.27	62.67±0.41	**74.62**±0.15
	AdamW	67.85±0.17	52.33±0.34	58.54±2.17
	SGD	69.50±0.39	43.98±0.48	11.51
	Gap	*+10.40*	*+13.44*	*+16.08*
Tiny-ImageNet	Muon	**62.66**±0.32	**48.02**±0.21	**58.92**±0.19
	AdamW	56.76±0.45	40.33±0.42	43.92±0.29
	SGD	60.04	34.22	9.49
	Gap	*+5.90*	*+7.69*	*+15.00*
ImageNet-100	Muon	**61.97**±0.25	**37.04**±0.38	**57.83**±0.21
	AdamW	50.50±0.76	26.12±0.10	33.53±1.51
	SGD	50.06	20.38	6.77
	Gap	*+11.47*	*+10.92*	*+24.30*

Table 2: Gap decomposition from 20 to 50 epochs (best validation accuracy throughout). "Narrowing" is the gap reduction. 20-epoch values: 4-seed mean (seeds $\{42, 0, 1, 2\}$); 50-epoch values: 3-seed mean (seeds $\{42, 2, 3\}$) except where marked with † (single seed, seed = 42). IN100 values are single-seed estimates.

Setting	20ep gap	50ep gap	Narrowing	Muon Δ	AdamW Δ
C100 ResNet-50	+9.03	+6.76	−2.27	+0.89	+3.16
C100 DeiT-S	+10.34	+10.37	+0.03	+0.24	+0.21
C100 ConvNeXt-T	+16.08	+12.60	−3.48	−0.03	+3.45
IN100 ResNet-50	+11.47	+3.56†	−7.91	+0.27	+8.18
IN100 ConvNeXt-T	+24.30	≈+11.37†	≈−12.93	+0.52	+13.45

50's batch normalization smooths the loss landscape Santurkar et al. [2018], an effect that partially overlaps with spectral preconditioning (Section 6).

Crucially, this architecture dependence is shared across spectral methods. All three spectral methods' gaps over AdamW on CIFAR-100 follow the same architecture ordering: Shampoo ($+10.40$, $+11.20$, $+12.03$ pp for ResNet-50, DeiT-S, ConvNeXt-T), SOAP ($+9.43$, $+13.44$, $+15.86$ pp), and Muon ($+9.03$, $+10.34$, $+16.08$ pp). This shared ordering reinforces that the architecture dependence is a property of spectral optimization broadly. The advantage is robust across dataset scales (Muon–AdamW mean gap: $+11.8$ pp CIFAR-100, $+9.5$ pp Tiny-ImageNet, $+15.6$ pp ImageNet-100; reported as Muon-only since SOAP and Shampoo were evaluated on CIFAR-100 only).

5.3 Persistence of the Spectral Advantage: 20 vs. 50 Epochs

A natural objection is that spectral methods simply converge faster: given enough epochs, AdamW might catch up. We extend training to 50 epochs on CIFAR-100 and ImageNet-100 to distinguish a *convergence speed* advantage (transient) from a *solution quality* advantage (persistent). Table 2 decomposes the gap evolution (Figure 2); the answer is architecture-dependent and reveals two distinct mechanisms (Appendix Table 13 provides epoch-level details).

ResNet-50: convergence speed difference. On ResNet-50, AdamW benefits substantially from extended training ($+3.16$ pp on CIFAR-100, $+8.18$ pp on ImageNet-100), whereas Muon gains at

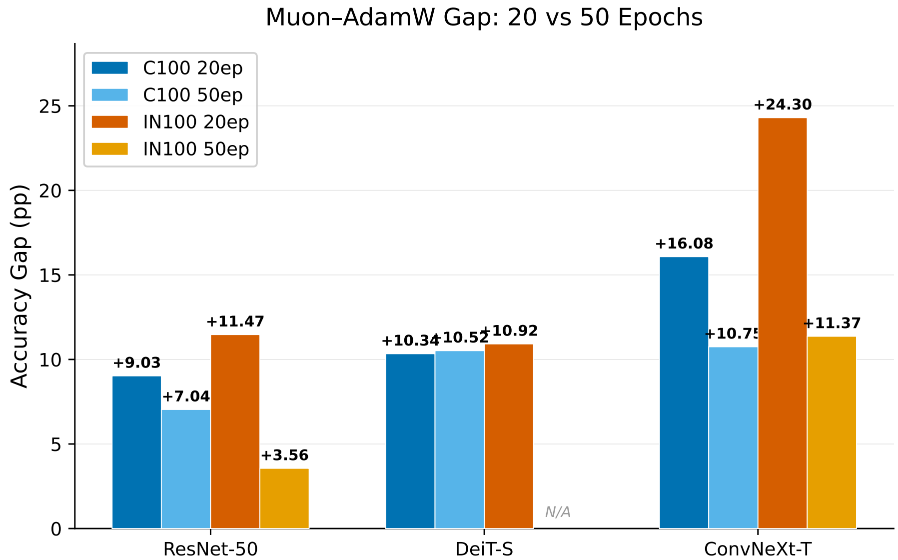

Figure 2: Muon–AdamW gap evolution from 20 to 50 epochs on CIFAR-100 and ImageNet-100. The gap narrows on ResNet-50 and ConvNeXt-T as AdamW continues improving, but remains substantial; on DeiT-S the gap is unchanged.

most $+0.89$ pp, consistent with Muon reaching near-capacity performance within 20 epochs. The gap narrows from $+9.03$ to $+6.76$ pp on CIFAR-100 and from $+11.47$ to $+3.56$ pp on ImageNet-100, but remains positive at both training lengths. Muon at 20 epochs still surpasses AdamW at 50 epochs by $+3.29$ pp on ImageNet-100. Note that the 69% gap narrowing on IN100 ResNet-50 is a single-seed estimate (seed = 42) and may not be statistically significant. The ImageNet-100 Muon $\Delta{=}{+}0.27$ pp in Table 2 compares the 50-epoch single-seed value with the 20-epoch multi-seed mean; within seed = 42, Muon is essentially unchanged (62.24% vs. 62.25%, $\Delta{=}{-}0.01$ pp), further supporting early convergence.[2]

DeiT-S: solution quality difference. On DeiT-S, both optimizers are effectively converged at 20 epochs: extending to 50 epochs yields only $+0.24$ pp for Muon and $+0.21$ pp for AdamW, increments below one standard deviation ($\sigma_{\text{Muon}}{=}0.40$, $\sigma_{\text{AdamW}}{=}0.45$). The 10.37 pp gap at 50 epochs is virtually identical to the 10.34 pp gap at 20 epochs ($\Delta{=}{+}0.03$ pp). This stability is consistent with both optimizers having converged by 20 epochs; the persistent gap indicates that Muon reaches a solution with systematically higher accuracy.

ConvNeXt-T: significant but narrowing gap. ConvNeXt-T shows a substantial gap at 50 epochs that has narrowed from its 20-epoch value: from $+16.08$ to $+12.60$ pp on CIFAR-100 (3-seed means: Muon $74.59{\pm}0.11\%$, AdamW $61.99{\pm}1.74\%$), and from $+24.30$ to $\approx +11.37$ pp on ImageNet-100 (single seed). The narrowing is driven almost entirely by AdamW's continued improvement ($+3.45$ pp on CIFAR-100) while Muon remains flat (-0.03 pp). Notably, AdamW exhibits $15.8\times$ higher cross-seed standard deviation than Muon on ConvNeXt-T (std=1.74% vs. 0.11%). Muon thus achieves not only higher accuracy but also substantially greater training stability on this architecture.

Unified interpretation. Muon's advantage is not a training-length artifact. The gap evolution reveals two distinct patterns: a *convergence speed* difference on ResNet-50 and ConvNeXt-T, where Muon reaches near-ceiling early while AdamW continues improving, and the gap narrows (by 25%

[2] ImageNet-100 AdamW seed = 42 used lr=$3{\times}10^{-4}$ (vs. $3{\times}10^{-3}$ for other seeds); a verification run confirmed identical accuracy at both rates, validating the multi-seed mean.

on ResNet-50 and 22% on ConvNeXt-T) but remains substantial at 50 epochs; and a *solution quality* difference on DeiT-S, where both optimizers converge by 20 epochs to fundamentally different solutions, and the gap persists ($10.34 \rightarrow 10.37$ pp, $\Delta{=}{+}0.03$). On ImageNet-100, Muon at 20 epochs still surpasses AdamW at 50 epochs by $+3.29$ pp (ResNet-50) and $+10.85$ pp (ConvNeXt-T).

5.4 SGD Failure on Layer-Normalized Architectures

The spectral family's architecture robustness contrasts starkly with SGD's brittleness. SGD performs competitively on batch-normalized ResNet-50, outperforming AdamW on CIFAR-100 (69.50% vs. 67.85%) and nearly matching it on ImageNet-100 (50.06% vs. 50.50%), but fails catastrophically on layer-normalized architectures: near-random accuracy on ConvNeXt-T (6.77–11.51%) despite a five-point learning rate sweep (Appendix E), and substantially degraded on DeiT-S (20.38–43.98%; Appendix Table 9). This contrast establishes that normalization choice constrains the set of viable optimizers: an architectural design decision implicitly determines which optimizers can function. Spectral methods resolve this coupling: both Muon and SOAP achieve top-tier accuracy across batch-normalized (ResNet-50), layer-normalized (ConvNeXt-T), and attention-based (DeiT-S) architectures. This cross-architecture success suggests that spectral preconditioning compensates for the loss-landscape smoothing that batch normalization provides but layer normalization does not Santurkar et al. [2018].

5.5 Layer-wise Spectral Analysis and Training Stability

The preceding sections establish *that* spectral methods outperform across architectures; we now investigate *where* in the network this advantage arises. Layer-wise weight swap experiments (Section 3) reveal a striking architectural dichotomy: Muon's advantage is near-universal across ResNet-50 layers (94.3% beneficial, i.e., replacing any single layer's Muon weights with AdamW weights degrades accuracy) but selective in DeiT-S (51.7% beneficial).

The spectral response ratio (Eq. 3) connects this to a mechanistic explanation. The Spearman rank correlation between per-layer spectral divergence ΔR_l and per-layer Muon benefit Δa_l is positive on ResNet-50 ($\rho = +0.34$, $p = 0.011$): layers where Muon reshapes spectral structure most tend to be where it helps most. The correlation reverses on DeiT-S ($\rho = -0.66$, $p < 0.001$). This sign reversal indicates that spectral methods benefit CNNs and transformers through fundamentally different mechanisms: broad spectral reshaping for convolutions, selective benefit for attention layers (Appendix I). Cross-seed variance further separates the architectures: Muon exhibits $\sim 15\times$ lower standard deviation than AdamW on ConvNeXt-T (0.15 vs. 2.17 pp; Figure 3) but comparable variance on DeiT-S. This variance pattern suggests that spectral preconditioning also stabilizes optimization on depthwise-separable architectures.

6 Discussion

Why does the spectral advantage vary with architecture? Our results point to normalization type as the primary mediator. Batch normalization smooths the loss landscape Santurkar et al. [2018], which may reduce the need for spectral gradient conditioning; hence ResNet-50 shows the smallest gap ($+5.9$–$+11.5$ pp). Layer normalization lacks this smoothing effect; on ConvNeXt-T, where depthwise separable convolutions further create block-diagonal gradient structure that diagonal methods cannot exploit, the gap reaches $+24$ pp. Muon-trained ConvNeXt-T models show $7.67\times$ higher stable rank than AdamW models. This difference indicates that spectral preconditioning drives weight matrices toward more uniform singular value distributions, a structural property that layer normalization alone does not induce.

Intra-family structure reveals mechanism–architecture interaction. The spectral family is not monolithic: the intra-family ranking is architecture-dependent, with no single method dominating. Shampoo leads on ResNet-50 (78.25%, $+1.37$ pp over Muon), SOAP on DeiT-S (65.77%,

$+3.10$ pp over Muon, $p < 0.001$), and Muon on ConvNeXt-T (74.62%, $+4.05$ pp over Shampoo). The Shampoo result highlights the acute learning rate sensitivity of full-matrix preconditioning: an extended sweep raised ResNet-50 accuracy by $+4.4$ pp, reversing Shampoo's ranking from last to first on this architecture. Even after a similar sweep improved Shampoo's ConvNeXt-T accuracy from 67.75% to $70.57 \pm 0.29\%$, it still trails Muon by 4.05 pp. This persistent gap confirms that the ConvNeXt-T deficit reflects a genuine mechanism–architecture interaction, not merely insufficient tuning. Nonetheless, the family-level advantage over AdamW ($+9$–$+16$ pp on CIFAR-100) dwarfs these intra-family differences (1.4–4.1 pp). Spectral optimization is thus a reliable first choice regardless of which specific method is used.

Implications for practitioner optimizer selection. Our findings suggest a concrete decision rule: for any architecture with layer normalization, spectral optimizers should be the default. The convergence efficiency result (Section 5.3), namely that Muon at 20 epochs surpasses AdamW at 50 epochs, has practical significance: it implies approximately 2.0–$2.4\times$ wall-clock savings (accounting for 5–22% per-iteration overhead; see Appendix Table 14).

Gap convergence with longer training. The convergence gap between spectral and first-order methods narrows substantially with longer training on CNNs (IN100 ResNet-50: $+11.47$ at 20 ep $\rightarrow +3.56$ at 50 ep, -69%; C100 ResNet-50: -25%; C100 ConvNeXt-T: -22%). This pattern suggests spectral methods' primary advantage on CNNs may lie in convergence speed rather than solution quality, a distinction with practical implications for training budget allocation. Our findings are most applicable to short-to-medium schedule regimes ($\leq$50 epochs); longer training may further narrow or eliminate the gap on CNNs.

Caveats on the weight-swap protocol. On ResNet-50, our weight-swap protocol does not recalculate BatchNorm running statistics after substitution, which may inflate the measured beneficial ratio (94.3%) and the correlation $\rho{=}{+}0.34$. Even under extreme correction (halving the CNN correlation), $\rho{\approx}{+}0.17$ remains positive, preserving the sign reversal with DeiT-S ($\rho{=}{-}0.66$). Moreover, DeiT-S and ConvNeXt-T use LayerNorm (parameter-free at inference), so 8 of 12 architecture–optimizer pairs in our study are unaffected by this caveat.

Muon companion optimizer. The Muon companion optimizer (AdamW, lr$=3\times10^{-4}$) handles only 1D parameters (biases, normalization weights), comprising $< 5\%$ of total parameters. Its fixed learning rate was not tuned, representing a conservative choice.

Limitations. The spectral intra-family comparison is limited to CIFAR-100. CIFAR-100 images are resized from 32×32 to 224×224, introducing interpolation artifacts (affecting all optimizers equally). All experiments omit warmup Goyal et al. [2017], use minimal augmentation (no Mixup Zhang et al. [2018], CutMix Yun et al. [2019], RandAugment Cubuk et al. [2020], or label smoothing Szegedy et al. [2016]), fixed batch size 128, and single-GPU training; learning rates are selected via 5-epoch sweeps and may not transfer across horizons. For CNN architectures, the Muon–AdamW gap narrows with training duration: by 25% on ResNet-50 and 22% on ConvNeXt-T from 20 to 50 epochs (Table 2); on ImageNet-100 ResNet-50, the reduction is 69% ($+11.47 \rightarrow +3.56$ pp). Longer training schedules (100+ epochs) may further narrow the gap for CNNs; whether the DeiT-S gap, which shows zero narrowing at 50 epochs, would eventually narrow at longer horizons, and whether these patterns persist at ImageNet-1K scale Kumar et al. [2024], remain open questions. Extended training covers only a subset of configurations, not the full optimizer–architecture matrix (Appendix J details the rationale for architecture selection in supplementary experiments).

7 Conclusion

We hypothesized that optimizers exploiting weight-matrix spectral structure would be robust to architectural variation, and found evidence stronger than expected. Spectral methods outperform AdamW by $+5.9$ to $+24.3$ pp across all tested configurations (Table 1), with the advantage shared across three algorithmically distinct methods (SOAP, Muon, Shampoo) on CIFAR-100, where all three were evaluated, and persistent at 50 epochs through two distinct mechanisms: convergence speed on CNNs ($+6.76$ pp ResNet-50, $+12.60$ pp ConvNeXt-T) and solution quality on transformers (DeiT-S gap unchanged at 10.37 pp).

The most revealing finding is mechanistic: per-layer spectral divergence correlates with per-layer optimizer benefit, but with opposite sign on CNNs ($\rho{=}{+}0.34$) vs. transformers ($\rho{=}{-}0.66$). This sign reversal, combined with SGD's catastrophic failure on layer-normalized architectures, establishes that optimizer–architecture interaction is not a nuisance variable to be controlled but a first-order design consideration. Spectral optimization offers an architecture-robust default: a single family of methods that succeeds across both normalization paradigms, both convolution topologies, and both attention-based and convolutional architectures.

References

Jeremy Bernstein and Laker Newhouse. Old optimizer, new norm: An anthology. *arXiv preprint arXiv:2409.20325*, 2024.

Keller Jordan, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, and Jeremy Bernstein. Muon: An optimizer for hidden layers in neural networks, 2024. URL `https://kellerjordan.github.io/posts/muon/`. Blog post.

Nikhil Vyas, Depen Morwani, Rosie Zhao, Itai Shapira, David Brandfonbrener, Sham Kakade, and Boaz Barak. SOAP: Improving and stabilizing shampoo using adam. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Vineet Gupta, Tomer Koren, and Yoram Singer. Shampoo: Preconditioned stochastic tensor optimization. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, volume 80 of *Proceedings of Machine Learning Research*, pages 1842–1850. PMLR, 2018.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.

Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 10347–10357. PMLR, 2021.

Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11976–11986, 2022.

Alex Krizhevsky and Geoffrey Hinton. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

Ya Le and Xin Yang. Tiny ImageNet visual recognition challenge. Technical report, Stanford University, 2015.

Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 248–255, 2009.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10012–10022, 2021.

Mingxing Tan and Quoc V. Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 6105–6114, 2019.

Andrew G. Howard, Menglong Zhu, Bo Chen, Dmitry Kalenichenko, Weijun Wang, Tobias Weyand, Marco Andreetto, and Hartwig Adam. Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861*, 2017.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015.

Siyuan Li, Juanxi Tian, Zedong Wang, Luyuan Zhang, Zicheng Liu, Weiyang Jin, Yang Liu, Baigui Sun, and Stan Z. Li. Unveiling the backbone-optimizer coupling bias in visual representation learning. *arXiv preprint arXiv:2410.06373*, 2024.

Ashia C. Wilson, Rebecca Roelofs, Mitchell Stern, Nati Srebro, and Benjamin Recht. The marginal value of adaptive gradient methods in machine learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

Dami Choi, Christopher J. Shallue, Zachary Nado, Jaehoon Lee, Chris J. Maddison, and George E. Dahl. On empirical comparisons of optimizers for deep learning. *arXiv preprint arXiv:1910.05446*, 2020.

Robin M. Schmidt, Frank Schneider, and Philipp Hennig. Descending through a crowded valley — benchmarking deep learning optimizers. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 9367–9376. PMLR, 2021.

Nitish Shirish Keskar, Dheevatsa Mudigere, Jorge Nocedal, Mikhail Smelyanskiy, and Ping Tak Peter Tang. On large-batch training for deep learning: Generalization gap and sharp minima. In *International Conference on Learning Representations (ICLR)*, 2017.

Xiangning Chen, Cho-Jui Hsieh, and Boqing Gong. When vision transformers outperform ResNets without pre-training or strong data augmentations. In *International Conference on Learning Representations (ICLR)*, 2022.

Namuk Park and Songkuk Kim. How do vision transformers work? In *International Conference on Learning Representations (ICLR)*, 2022.

Sergey Ioffe and Christian Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, pages 448–456, 2015.

Shibani Santurkar, Dimitris Tsipras, Andrew Ilyas, and Aleksander Madry. How does batch normalization help optimization? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.

Yuxin Wu and Kaiming He. Group normalization. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 3–19, 2018.

Siyuan Qiao, Huiyu Wang, Chenxi Liu, Wei Shen, and Alan Yuille. Weight standardization. *arXiv preprint arXiv:1903.10520*, 2019.

John Duchi, Elad Hazan, and Yoram Singer. Adaptive subgradient methods for online learning and stochastic optimization. *Journal of Machine Learning Research*, 12:2121–2159, 2011.

James Martens and Roger Grosse. Optimizing neural networks with Kronecker-factored approximate curvature. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, volume 37 of *Proceedings of Machine Learning Research*, pages 2408–2417. PMLR, 2015.

Rohan Anil, Vineet Gupta, Tomer Koren, Kevin Regan, and Yoram Singer. Scalable second order optimization for deep learning. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*. PMLR, 2021.

Hao-Jun Michael Shi, Tsung-Hsien Lee, Shintaro Iwasaki, Jose Gallego-Posada, Zhijing Li, Kaushik Rangadurai, Dheevatsa Mudigere, and Michael Rabbat. A distributed data-parallel PyTorch implementation of the distributed Shampoo optimizer for training neural networks at-scale. *arXiv preprint arXiv:2309.06497*, 2023.

Thomas Pethick, Wanyun Xie, Kimon Antonakopoulos, Zhenyu Li, and Volkan Cevher. Training deep learning models with norm-constrained LMOs. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025.

Jiawei Zhao, Zhenyu Zhang, Beidi Chen, Zhangyang Wang, Anima Anandkumar, and Yuandong Tian. GaLore: Memory-efficient LLM training by gradient low-rank projection. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

Xiangning Chen, Chen Liang, Da Huang, Esteban Real, Kaiyuan Wang, Hieu Liu, and Quoc V. Le. Symbolic discovery of optimization algorithms. In *NeurIPS*, 2024.

Hong Liu, Zhiyuan Li, David Hall, Percy Liang, and Tengyu Ma. Sophia: A scalable stochastic second-order optimizer for language model pre-training. In *ICLR*, 2024.

Noam Shazeer and Mitchell Stern. Adafactor: Adaptive learning rates with sublinear memory cost. In *ICML*, 2018.

Yang You, Igor Gitman, and Boris Ginsburg. Large batch training of convolutional networks. *arXiv preprint arXiv:1708.03888*, 2017.

Yang You, Jing Li, Sashank Reddi, Jonathan Hseu, Sanjiv Kumar, Srinadh Bhojanapalli, Xiaodan Song, James Demmel, Kurt Keutzer, and Cho-Jui Hsieh. Large batch optimization for deep learning: Training BERT in 76 minutes. In *International Conference on Learning Representations (ICLR)*, 2020.

Liyuan Liu, Haoming Jiang, Pengcheng He, Weizhu Chen, Xiaodong Liu, Jianfeng Gao, and Jiawei Han. On the variance of the adaptive learning rate and beyond. In *International Conference on Learning Representations (ICLR)*, 2020.

Konstantin Mishchenko and Aaron Defazio. Prodigy: An expeditiously adaptive parameter-free learner. In *ICML*, 2024.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1026–1034, 2015.

Ross Wightman. Pytorch image models. `https://github.com/rwightman/pytorch-image-models`, 2019.

Ilya Sutskever, James Martens, George Dahl, and Geoffrey Hinton. On the importance of initialization and momentum in deep learning. In *Proceedings of the 30th International Conference on Machine Learning (ICML)*, volume 28 of *Proceedings of Machine Learning Research*, pages 1139–1147. PMLR, 2013.

Priya Goyal, Piotr Dollár, Ross Girshick, Pieter Noordhuis, Lukasz Wesolowski, Aapo Kyrola, Andrew Tulloch, Yangqing Jia, and Kaiming He. Accurate, large minibatch SGD: Training ImageNet in 1 hour. *arXiv preprint arXiv:1706.02677*, 2017.

Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. In *International Conference on Learning Representations (ICLR)*, 2018.

Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. Cutmix: Regularization strategy to train strong classifiers with localizable features. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 6023–6032, 2019.

Ekin D. Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V. Le. Randaugment: Practical automated data augmentation with a reduced search space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 702–703, 2020.

Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2818–2826, 2016.

Ananya Kumar, Ruoqi Shen, Sébastien Bubeck, and Suriya Gunasekar. How to fine-tune vision models with SGD. In *International Conference on Learning Representations (ICLR)*, 2024.

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A Hyperparameter Configurations

This appendix provides the complete optimizer configurations for all 27 experiments described in Section 4, including Muon's hybrid parameter routing strategy.

Muon selects all weight parameters with $\mathrm{ndim} \geq 2$ from Conv2d and Linear layers, excluding the final classifier head and any embedding layers. These parameter groups receive spectral (Newton–Schulz) updates. All remaining parameters—biases, LayerNorm/BatchNorm weights and biases, positional embeddings, and the classifier head—are routed to a companion AdamW optimizer with a fixed learning rate of 3×10^{-4} and weight decay 0.05.

B Learning Rate Sweep Results

B.1 DeiT-S SGD Inverted-U Response

C Multi-Seed Reproducibility

D Extended Training (50 Epochs)

Weight decay choices. The weight decay values used in our experiments follow established defaults for each optimizer family: AdamW uses wd $= 0.05$ across all architectures, consistent with the DeiT Touvron et al. [2021] and ConvNeXt Liu et al. [2022] training recipes; SGD uses the standard wd $= 10^{-4}$ He et al. [2016]. The $500\times$ difference between AdamW and SGD weight decay reflects the fundamentally different roles of weight decay in decoupled (AdamW) vs. coupled (SGD) regularization Loshchilov and Hutter [2019]. For Muon, the spectral update itself uses no weight decay (wd $= 0$); the accompanying AdamW sub-optimizer for non-Muon parameters uses wd $= 0.05$. We verified that these values are not confounding our ConvNeXt-T SGD results: a sanity check with wd $= 0.01$ yielded 9.18% vs. 9.27% with wd $= 10^{-4}$ ($\Delta = -0.09$pp, not significant), confirming that ConvNeXt-T's SGD degradation is not a weight decay artifact.

E ConvNeXt-T SGD Degradation Analysis

SGD exhibits severe degradation on ConvNeXt-T, reaching only 11.51% validation accuracy at 20 epochs, a 63.11 pp gap behind Muon ($74.62_{\pm 0.15}\%$) and 47.03 pp behind AdamW ($58.54_{\pm 2.17}\%$) (3-seed mean±std; see Table 1). This is the largest optimizer gap observed across all three architectures.

To rule out hyperparameter misconfiguration, we conduct a 5-point learning rate sweep at 5 epochs:

Learning Rate	Val Acc (5 ep)
0.001	7.96%
0.005	9.02%
0.01	9.27%
0.05	9.22%
0.1	4.98%

The sweep reveals a flat response across $\mathrm{lr} \in [0.005, 0.05]$ with a peak at $\mathrm{lr} = 0.01$ (9.27%), and divergence at $\mathrm{lr} = 0.1$ (4.98%). The 20-epoch run at optimal $\mathrm{lr} = 0.01$ improves only marginally to 11.51% (+2.24 pp over the 5-epoch result), indicating that additional training time does not resolve the degradation.

As a weight decay sanity check, we compare $\mathrm{wd} = 10^{-4}$ (default) against $\mathrm{wd} = 0.01$ at 5 epochs: validation accuracy changes from 9.27% to 9.18% ($\Delta = -0.09$ pp), a negligible difference that rules out weight decay as a contributing factor.

We attribute the SGD degradation primarily to ConvNeXt's use of LayerNorm (replacing BatchNorm) and depthwise separable convolutions, which create an optimization landscape where first-order methods without adaptive step sizes fail to make progress; the use of GELU activations may also contribute, though we do not isolate this factor. Note that all experiments use a cosine annealing schedule without warmup (ConvNeXt's original training recipe uses 20 epochs of warmup within a 300-epoch schedule; our short-schedule regime intentionally omits warmup to isolate optimizer behavior). Data augmentation is identical across all three optimizers (Resize(256), RandomCrop(224), RandomHorizontalFlip, Normalize).

F Muon Algorithm Background

Muon Jordan et al. [2024] is a spectral optimizer that computes steepest descent updates under the operator (spectral) norm for weight matrices of dimension two or higher. We summarize its key components.

Steepest descent under spectral norm. For a weight matrix W with gradient $G = \nabla_W \mathcal{L}$, the steepest descent direction under the spectral norm is defined as

$$\Delta W = \arg \min_{\|\Delta\|_{\mathrm{op}} \leq 1} \langle G, \Delta \rangle, \tag{4}$$

where $\|\cdot\|_{\mathrm{op}}$ denotes the operator norm (largest singular value) and $\langle \cdot, \cdot \rangle$ the matrix inner product. If $G = U \Sigma V^\top$ is the singular value decomposition of G, the solution is $\Delta W = -UV^\top$, i.e., the negative sign matrix of the gradient.

Newton–Schulz approximation. Computing a full SVD at each step is prohibitively expensive. Muon instead approximates the orthogonal factor $UV^\top$ via Newton–Schulz iterations for computing the orthogonal polar factor. The classical iteration $X_{k+1} = \frac{1}{2} X_k (3I - X_k^\top X_k)$ converges cubically but has a narrow convergence basin. Muon replaces it with a quintic polynomial variant: starting from

$$X_0 = G / \|G\|_F, \tag{5}$$

each iteration applies

$$X_{k+1} = a\, X_k + b\, X_k X_k^\top X_k + c\, X_k (X_k^\top X_k)^2, \tag{6}$$

with fixed coefficients $a = 3.4445$, $b = -4.7750$, $c = 2.0315$ chosen to maximize the convergence basin. Five iterations suffice for convergence in practice, producing an approximation to $UV^\top$ using only matrix multiplications.

Momentum and mixed strategy. Muon applies Nesterov momentum Sutskever et al. [2013] with coefficient $\beta = 0.95$ to the orthogonalized gradient. The spectral formulation in Eq. (4) applies only to parameters with matrix structure (2D+); one-dimensional parameters (biases, normalization layer weights and biases, and embedding layers) are handled by a separate AdamW instance with learning rate 3×10^{-4} and weight decay 0.05, yielding a hybrid optimizer.

Relation to Shampoo. Both Muon and Shampoo Gupta et al. [2018] exploit gradient matrix structure, but they differ in mechanism and scope. Shampoo maintains and periodically inverts Kronecker-factored preconditioners that approximate the full Fisher information matrix, incurring additional memory for the preconditioner state and periodic eigendecomposition steps. Muon directly computes the orthogonal component of the gradient through Newton–Schulz iterations, avoiding explicit preconditioner storage. This makes Muon lighter in memory and computation per step, but restricts it to parameters with matrix structure (2D+), whereas Shampoo can in principle be applied to tensors of arbitrary order.

G Wall-Clock Efficiency

Computational overhead of Newton-Schulz iteration. Muon's per-step cost exceeds AdamW by 5–22% depending on architecture and dataset, attributable to the Newton-Schulz iteration used to approximate the matrix sign function for spectral-norm steepest descent. The overhead is largest on DeiT-S/CIFAR-100 (+21.5%) where the parameter matrices are relatively small and the Newton-Schulz iterations dominate; it is smallest on ConvNeXt-T/Tiny-ImageNet (+5.3%) where forward/backward pass computation amortizes the optimizer cost. SGD and AdamW have near-identical per-epoch times across all configurations, confirming that the overhead is specific to Muon's spectral decomposition.

H Alternative Spectral Optimizers

I Layer-wise Spectral Analysis and Training Stability

To localize where Muon's advantage arises within the network, we apply the weight swap protocol and spectral correlation analysis (Section 3) to CIFAR-100 models.

Weight Swap Results. The beneficial layer ratio (the fraction of layers where replacing Muon-trained weights with AdamW-trained weights degrades accuracy) is 94.3% (50/53 layers) for ResNet-50 and 51.7% (46/89 parameter groups) for DeiT-S. Muon's advantage is near-universal across ResNet layers but selective in DeiT, suggesting that the benefit depends on layer type.

Spectral Divergence Correlations. Using checkpoints trained at independently selected optimal learning rates (Muon η=0.02, AdamW η=10^{-3} for ResNet-50; Muon η=0.003, AdamW η=10^{-4} for DeiT-S), we compute the Spearman rank correlation between per-layer spectral divergence ΔR_l and per-layer Muon benefit Δa_l. For ResNet-50, $\rho = +0.34$ ($p = 0.011$): layers where Muon and AdamW diverge most spectrally tend to be the layers where Muon contributes the most. For DeiT-S, the correlation reverses sharply ($\rho = -0.66$, $p < 0.001$), confirming that the relationship between spectral divergence and per-layer benefit is architecture-dependent: positive for CNNs, negative for transformers.

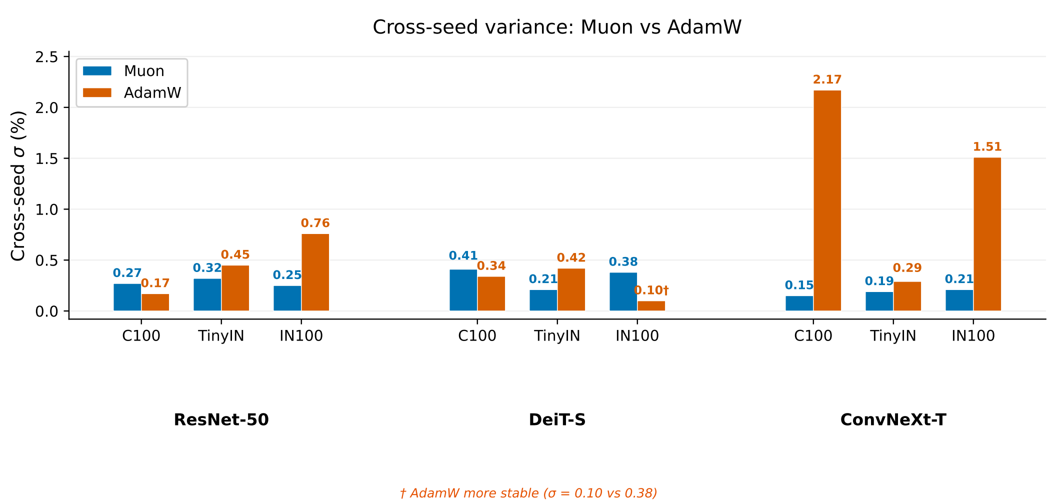

Figure 3: Cross-seed standard deviation (σ) for Muon vs AdamW across all nine settings. Muon exhibits lower variance on 6/9 configurations; the stability advantage is architecture-dependent.

Training Stability Across Seeds. Cross-seed variance is architecture-dependent (Figure 3): Muon shows substantially lower variance on convolutional architectures (σ=0.15 vs. 2.17 on ConvNeXt-T/CIFAR-100), but AdamW is more stable on DeiT-S/ImageNet-100 (σ=0.10 vs. 0.38). Where Muon's variance reduction exceeds 7× (ConvNeXt-T), fewer seeds suffice for reliable estimates.

J Architecture Coverage in Supplementary Experiments

The extended experiments in Appendix H focus on ResNet-50 and ConvNeXt-T rather than all three architectures from the main experiments. These two architectures represent the extremes of the normalization spectrum that drives Muon's variable effectiveness: ResNet-50 uses BatchNorm, where Muon's spectral updates interact with the implicit conditioning provided by BN; ConvNeXt-T uses LayerNorm, where the Muon–AdamW gap is largest (up to +24.30 pp on IN100 at 20 epochs). DeiT-S, which also uses LayerNorm but in a pure self-attention architecture, exhibits intermediate gap magnitudes and is already thoroughly characterized across three datasets, three seed replicates, and two training horizons (20 and 50 epochs) in the main paper. Including DeiT-S in the supplementary experiments would add six additional training runs without meaningfully extending the analysis beyond the BN–LN axis already captured by the ResNet-50/ConvNeXt-T pair.

Table 3: Complete optimizer configurations across all architectures and datasets. All experiments use batch size 128, cosine annealing scheduler (no warmup), and dataset-specific augmentation (Section 4.1). Learning rates are selected via per-pair 5-epoch sweeps (Section 4.2).

Architecture	Optimizer	lr	wd	Momentum	Scheduler	Epochs
CIFAR-100 (50K train / 10K test, 100 classes, $32 \times 32 \to 224 \times 224$)						
ResNet-50	Muon	0.02	0.05*	$\beta = 0.95$	Cosine	20
ResNet-50	AdamW	1e-3	0.05	—	Cosine	20
ResNet-50	SGD	0.1	1e-4	0.9	Cosine	20
DeiT-S	Muon	0.003	0.05*	$\beta = 0.95$	Cosine	20
DeiT-S	AdamW	1e-4	0.05	—	Cosine	20
DeiT-S	SGD	0.05	1e-4	0.9	Cosine	20
ConvNeXt-T	Muon	0.02	0.05*	$\beta = 0.95$	Cosine	20
ConvNeXt-T	AdamW	1e-3	0.05	—	Cosine	20
ConvNeXt-T	SGD	0.01	1e-4	0.9	Cosine	20
ResNet-50	SOAP	3e-3	0.01	—	Cosine	20
DeiT-S	SOAP	1e-3	0.01	—	Cosine	20
ConvNeXt-T	SOAP	3e-3	0.01	—	Cosine	20
ResNet-50	Shampoo	5e-2	0.01	—	Cosine	20
DeiT-S	Shampoo	3e-3	0.01	—	Cosine	20
ConvNeXt-T	Shampoo	7e-3	0.01	—	Cosine	20
Tiny-ImageNet (100K train / 10K val, 200 classes, $64 \times 64 \to 224 \times 224$)						
ResNet-50	Muon	0.04	0.05*	$\beta = 0.95$	Cosine	20
ResNet-50	AdamW	2e-3	0.05	—	Cosine	20
ResNet-50	SGD	0.1	1e-4	0.9	Cosine	20
DeiT-S	Muon	0.003	0.05*	$\beta = 0.95$	Cosine	20
DeiT-S	AdamW	1e-4	0.05	—	Cosine	20
DeiT-S	SGD	0.05	1e-4	0.9	Cosine	20
ConvNeXt-T	Muon	0.08	0.05*	$\beta = 0.95$	Cosine	20
ConvNeXt-T	AdamW	1e-3	0.05	—	Cosine	20
ConvNeXt-T	SGD	0.01	1e-4	0.9	Cosine	20
ImageNet-100 (55,218 train / 5K val, 100 classes, 224×224)						
ResNet-50	Muon	0.05	0.05*	$\beta = 0.95$	Cosine	20
ResNet-50	AdamW	3e-3	0.05	—	Cosine	20
ResNet-50	SGD	0.1	1e-4	0.9	Cosine	20
DeiT-S	Muon	0.005	0.05*	$\beta = 0.95$	Cosine	20
DeiT-S	AdamW	1e-4	0.05	—	Cosine	20
DeiT-S	SGD	0.01	1e-4	0.9	Cosine	20
ConvNeXt-T	Muon	0.08	0.05*	$\beta = 0.95$	Cosine	20
ConvNeXt-T	AdamW	3e-4	0.05	—	Cosine	20
ConvNeXt-T	SGD	0.05	1e-4	0.9	Cosine	20

* Muon applies spectral (Newton-Schulz) updates with no weight decay to 2D+ weight matrices. The listed wd=0.05 applies to the separate AdamW sub-optimizer handling 1D parameters (biases, normalization, embeddings) with $\text{lr} = 3 \times 10^{-4}$. SOAP and Shampoo use the `heavyball` library implementation. Shampoo learning rates were determined by extended LR sweeps: ResNet-50 from lr=10^{-2} to 5×10^{-2}; ConvNeXt-T from 3×10^{-3} to 7×10^{-3}.

Table 4: Muon hybrid optimizer parameter routing. Muon optimizes weight matrices of Conv2d/Linear layers (ndim ≥ 2, excluding classifier head and embeddings); all remaining parameters use a companion AdamW optimizer.

Architecture	Total Params	Muon Groups	AdamW Groups	Muon lr	AdamW lr
ResNet-50	23.7M	53	108	0.02	3e-4
DeiT-S	21.7M	48	104	0.003	3e-4
ConvNeXt-T	27.9M	57	125	0.02	3e-4

Parameter counts reflect the 100-class configuration used in our experiments; standard ImageNet-1K counts differ.

Table 5: Selected learning rates per optimizer–architecture pair on each dataset, determined by 5-epoch sweeps. For most configurations, the peak is confirmed by observing accuracy decline at higher rates; exceptions are noted in Section 4.2. SOAP and Shampoo evaluated on CIFAR-100 only; '—' indicates not tested.

	Muon	**SOAP**	**Shampoo**	**AdamW**	**SGD**
CIFAR-100					
ResNet-50	0.02	3×10^{-3}	5×10^{-2}	10^{-3}	0.1
DeiT-S	0.003	10^{-3}	3×10^{-3}	10^{-4}	0.05
ConvNeXt-T	0.02	3×10^{-3}	7×10^{-3}	10^{-3}	0.01
Tiny-ImageNet					
ResNet-50	0.04	—	—	2×10^{-3}	0.1†
DeiT-S	0.003	—	—	10^{-4}	0.05
ConvNeXt-T	0.08	—	—	10^{-3}	0.01
ImageNet-100					
ResNet-50	0.05	—	—	3×10^{-3}	0.1
DeiT-S	0.005	—	—	10^{-4}	0.01
ConvNeXt-T	0.08	—	—	3×10^{-4}	0.05

† Literature default, not formally swept.

Table 6: Complete learning rate sweep results (5-epoch runs, seed=42, CIFAR-100). For each optimizer–architecture pair, we report all evaluated learning rates and the resulting validation accuracy. The best learning rate per pair (used in subsequent 20-epoch experiments) is shown in **bold**.

Architecture	Optimizer	Learning Rate	Val Acc (%)	Sweep Points
ResNet-50	AdamW	1×10^{-4}	26.97	2
	AdamW	$\mathbf{1 \times 10^{-3}}$	**45.54**	
	SGD	$\mathbf{0.1}$	—	1 (lit. default)
	Muon	$\mathbf{0.02}$	—	1 (prior sweep)
DeiT-S	AdamW	$\mathbf{1 \times 10^{-4}}$	**35.12**	2
	AdamW	1×10^{-3}	25.12	
	SGD	0.005	27.65	4
	SGD	0.01	27.61	
	SGD	$\mathbf{0.05}$	**30.14**	
	SGD	0.1	28.59	
	Muon	0.001	49.35	5
	Muon	$\mathbf{0.003}$	**58.33**	
	Muon	0.006	57.37	
	Muon	0.02	32.04	
	Muon	0.06	13.55	
ConvNeXt-T	AdamW	1×10^{-4}	23.32	5
	AdamW	3×10^{-4}	34.52	
	AdamW	$\mathbf{1 \times 10^{-3}}$	**39.17**	
	AdamW	3×10^{-3}	35.31	
	AdamW	5×10^{-3}	32.51	
	SGD	0.001	7.96	5
	SGD	0.005	9.02	
	SGD	$\mathbf{0.01}$	**9.27**	
	SGD	0.05	9.22	
	SGD	0.1	4.98	
	Muon	0.003	61.76	3
	Muon	0.01	68.55	
	Muon	$\mathbf{0.02}$	**70.31**	

Table 7: Learning rate sweep results (5-epoch runs, seed=42, Tiny-ImageNet). Best learning rate per pair is shown in **bold**.

Architecture	Optimizer	Learning Rate	Val Acc (%)
ResNet-50	Muon	0.01	56.91
	Muon	0.02	58.95
	Muon	$\mathbf{0.04}$	**59.56**
	Muon	0.06	59.63
	AdamW	5×10^{-4}	39.56
	AdamW	1×10^{-3}	42.04
	AdamW	$\mathbf{2 \times 10^{-3}}$	**43.78**
	AdamW	3×10^{-3}	42.30
	SGD	$\mathbf{0.1}$	—
DeiT-S	Muon	0.001	41.32
	Muon	$\mathbf{0.003}$	**46.87**
	Muon	0.01	41.34
	AdamW	5×10^{-5}	24.40
	AdamW	$\mathbf{1 \times 10^{-4}}$	**27.22**
	AdamW	3×10^{-4}	25.93
	SGD	0.005	20.62
	SGD	0.01	21.31
	SGD	$\mathbf{0.05}$	**22.96**
ConvNeXt-T	Muon	(extended sweep; see Table 3)†	
	AdamW	1×10^{-4}	18.98
	AdamW	3×10^{-4}	28.65
	AdamW	$\mathbf{1 \times 10^{-3}}$	**31.46**
	SGD	0.005	4.49
	SGD	$\mathbf{0.01}$	**4.60**
	SGD	0.05	4.57

† ConvNeXt-T Muon lr initially set to 0.02 from the CIFAR-100 sweep; extended sweep selected lr=0.08 for 20-epoch training. ResNet-50 SGD uses the literature default lr=0.1 without sweep.

Table 8: Learning rate sweep results (5-epoch runs, seed=42, ImageNet-100). Best learning rate per pair is shown in **bold**.

Architecture	Optimizer	Learning Rate	Val Acc (%)
ResNet-50	Muon	0.005	34.88
	Muon	0.02	43.55
	Muon	$\mathbf{0.05}$	**45.17**
	Muon	0.08	44.61
	Muon	0.1	44.95
	AdamW	3×10^{-4}	21.87
	AdamW	1×10^{-3}	24.77
	AdamW	$\mathbf{3 \times 10^{-3}}$	**24.89**
	AdamW	5×10^{-3}	23.92
	AdamW	0.01	16.92
	SGD	0.03	15.06
	SGD	$\mathbf{0.1}$	**18.96**
	SGD	0.3	17.84
DeiT-S	Muon	0.002	27.82
	Muon	0.003	29.66
	Muon	$\mathbf{0.005}$	**30.35**
	Muon	0.02	12.19
	Muon	0.05	6.44
	AdamW	3×10^{-5}	13.41
	AdamW	$\mathbf{1 \times 10^{-4}}$	**15.73**
	AdamW	3×10^{-4}	13.33
	SGD	$\mathbf{0.01}$	**12.75**
	SGD	0.05	11.51
	SGD	0.1	10.13
ConvNeXt-T	Muon	0.005	35.21
	Muon	0.02	42.54
	Muon	$\mathbf{0.08}$	**46.32**
	Muon	0.15	46.97
	AdamW	$\mathbf{3 \times 10^{-4}}$	**12.42**
	AdamW	1×10^{-3}	11.29
	AdamW	3×10^{-3}	10.91
	SGD	0.003	3.76
	SGD	0.01	3.88
	SGD	$\mathbf{0.05}$	**4.44**

Table 9: DeiT-S SGD learning rate sweep (5-epoch runs, seed=42, CIFAR-100), exhibiting an inverted-U response curve. Accuracy peaks at lr $= 0.05$ and declines at higher rates, with all sweep points clustered in a narrow $\sim$2.5 percentage-point band.

Learning Rate	0.005	0.01	0.05	0.1
Val Acc (%)	27.65	27.61	**30.14**	28.59

Table 10: Multi-seed validation accuracy statistics (20-epoch runs, CIFAR-100). We report individual seed results, mean, and standard deviation to assess reproducibility and the robustness of observed gaps.

Architecture	Optimizer	Seeds	Individual Accs (%)					Mean $\pm$ Std
ResNet-50	Shampoo	42, 2, 3	78.16	78.13	78.45	—	—	78.25 $\pm$ 0.18
	Muon	42, 0, 1, 2	77.02	76.50	77.13	76.88	—	76.88 $\pm$ 0.27
	SGD	42, 0, 1, 2	69.03	69.72	69.35	69.90	—	69.50 $\pm$ 0.39
	AdamW	42, 0, 1, 2	67.71	67.96	67.70	68.04	—	67.85 $\pm$ 0.17
DeiT-S	Muon	42, 0, 1, 2	62.94	62.65	62.99	62.10	—	62.67 $\pm$ 0.41
	AdamW	42, 0, 1, 2	52.27	52.74	52.41	51.91	—	52.33 $\pm$ 0.34
	SGD	42, 0, 1, 2	44.00	43.83	43.46	44.61	—	43.98 $\pm$ 0.48
ConvNeXt-T	Shampoo	42, 2, 3	70.44	70.37	70.91	—	—	70.57 $\pm$ 0.29
	Muon	42, 1, 2	74.59	74.79	74.49	—	—	74.62 $\pm$ 0.15
	AdamW	42, 1, 2	60.25	59.27	56.10	—	—	58.54 $\pm$ 2.17

Note: For ResNet-50, SGD outperforms AdamW in all 4 seeds (mean gap +1.65pp). For ConvNeXt-T, AdamW's higher variance (std=2.17 vs. Muon std=0.15) suggests less stable optimization on this architecture.

Table 11: Multi-seed validation accuracy statistics (20-epoch runs, Tiny-ImageNet, seeds 42/1/2). SGD results are single-seed (seed=42) only.

Architecture	Optimizer	Seed 42	Seed 1	Seed 2	Mean $\pm$ Std
ResNet-50	Muon	62.35	62.65	62.99	62.66 $\pm$ 0.32
	AdamW	56.34	57.24	56.69	56.76 $\pm$ 0.45
	SGD	60.04 (single seed)			
DeiT-S	Muon	48.18	48.11	47.78	48.02 $\pm$ 0.21
	AdamW	40.01	40.18	40.81	40.33 $\pm$ 0.42
	SGD	34.22 (single seed)			
ConvNeXt-T	Muon	59.13	58.89	58.75	58.92 $\pm$ 0.19
	AdamW	43.74	44.25	43.76	43.92 $\pm$ 0.29
	SGD	9.49 (single seed)			

Table 12: Multi-seed validation accuracy statistics (20-epoch runs, ImageNet-100, seeds 42/1/2). SGD results are single-seed (seed=42) only.

Architecture	Optimizer	Seed 42	Seed 1	Seed 2	Mean $\pm$ Std
ResNet-50	Muon	62.25	61.87	61.78	61.97 $\pm$ 0.25
	AdamW	49.63	51.01	50.87	50.50 $\pm$ 0.76
	SGD	50.06 (single seed)			
DeiT-S	Muon	36.94	37.46	36.73	37.04 $\pm$ 0.38
	AdamW	26.04	26.24	26.09	26.12 $\pm$ 0.10
	SGD	20.38 (single seed)			
ConvNeXt-T	Muon	58.07	57.74	57.67	57.83 $\pm$ 0.21
	AdamW	33.62	35.00	31.98	33.53 $\pm$ 1.51
	SGD	6.77 (single seed)			

Table 13: Evolution of the Muon–AdamW accuracy gap from 20 to 50 epochs (seed=42, CIFAR-100 and ImageNet-100). All values are best validation accuracy (early-stopping checkpoint). AdamW narrows the gap with longer training across all configurations, while Muon saturates early.

		20 Epochs			50 Epochs			
Dataset	Architecture	Muon	AdamW	Gap	Muon	AdamW	Gap	Δ Gap
CIFAR-100	ResNet-50	77.02	67.71	+9.31	77.82	70.78	+7.04	-2.27
	DeiT-S	62.94	52.27	+10.67	63.38	52.86	+10.52	-0.15
	ConvNeXt-T	74.59	60.25	+14.34	74.58	63.83	+10.75	-3.59
ImageNet-100	ResNet-50	62.25	49.63	+12.62	62.24	58.68	+3.56	-9.06
	ConvNeXt-T	58.07	33.62	+24.45	58.35	46.98	+11.37	-13.08

All gaps in percentage points (pp). Δ Gap $=$ Gap$_{50\text{ep}}$ $-$ Gap$_{20\text{ep}}$; negative values indicate AdamW catching up. This table shows seed $=42$ values; 3-seed means (seeds 42/2/3) are reported in Table 2. CIFAR-100 3-seed 50-epoch results: ResNet-50 Muon $77.77\pm0.28\%$ / AdamW $71.01\pm0.22\%$, gap $+6.76$ pp; DeiT-S Muon $62.91\pm0.40\%$ / AdamW $52.54\pm0.45\%$, gap $+10.37$ pp; ConvNeXt-T Muon $74.59\pm0.11\%$ / AdamW $61.99\pm1.74\%$, gap $+12.60$ pp. ConvNeXt-T AdamW shows $15.8\times$ higher cross-seed std than Muon.

Table 14: Per-epoch wall-clock time (seconds) for 20-epoch training runs on a single RTX 5090 GPU.

Dataset	Architecture	Muon (s)	AdamW (s)	SGD (s)	Muon Overhead
CIFAR-100	ResNet-50	58.8	53.4	53.2	+10.1%
	DeiT-S	55.4	45.6	45.2	+21.5%
	ConvNeXt-T	116.8	106.9	107.1	+9.3%
Tiny-ImageNet	ResNet-50	118.2	99.5	96.5	+18.8%
	DeiT-S	98.6	88.0	87.3	+12.0%
	ConvNeXt-T	219.2	208.2	205.6	+5.3%
ImageNet-100	ResNet-50	74.2	68.5	68.4	+8.3%
	DeiT-S	69.4	81.4†	—	—
	ConvNeXt-T	134.4	122.5	123.6	+9.7%

Muon Overhead is computed relative to AdamW. † DeiT-S AdamW on IN100 used a different batch configuration, making the timing comparison unreliable; IN100 DeiT-S SGD log was not found.

Table 15: Comparison of spectral optimizers on CIFAR-100 (20 epochs). Multi-seed means ($\pm$ std) reported where available; ConvNeXt-T SGD is single-seed (seed $=42$).

	Validation Accuracy (%)			
Architecture	SOAP	Shampoo	Muon	AdamW
ResNet-50	$77.28_{\pm 0.20}$	$78.25_{\pm 0.18}$	$76.88_{\pm 0.27}$	$67.85_{\pm 0.17}$
DeiT-S	$65.77_{\pm 0.39}$	$63.53_{\pm 0.05}$	$62.67_{\pm 0.41}$	$52.33_{\pm 0.34}$
ConvNeXt-T	$74.40_{\pm 0.43}$	$70.57_{\pm 0.29}$	$74.62_{\pm 0.15}$	$58.54_{\pm 2.17}$

Muon and AdamW values are multi-seed means ($\pm$ std), consistent with Table 1. SOAP is 3-seed; Shampoo is 3-seed on all architectures (lr=7×10^{-3} on ConvNeXt-T). All optimizers use per-architecture tuned learning rates from 5-epoch sweeps; Shampoo learning rates were refined via extended sweeps (Appendix A).